

- Q.27 State and explain laws of illumination. (CO1)
 Q.28 Explain constructional differences of energy efficient motor from normal motor. (CO4)
 Q.29 List steps involved in energy audit. (CO3)
 Q.30 State the significant features of energy efficient motors. (CO4)
 Q.31 Write a short note on amorphous core transformer. (CO4)
 Q.32 State factors affecting energy efficiency. (CO5)
 Q.33 Explain the benefits of power factor improvement for consumer as well as for generating stations. (CO5)
 Q.34 What is tariff? Why it is required to frame a tariff? (CO4)
 Q.35 Explain effects of low power factor? (CO5)

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
 Q.36 How energy can be saved using energy efficient devices? What is the need of energy efficient devices? (CO4)
 Q.37 Explain efficient pumping system operation. What are energy saving tips in pumps? (CO5)
 Q.38 a) Describe LED and its use. (CO1,3)
 b) Explain the duties of energy auditor and energy manager.

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6th Sem / Electrical Engg.

Sub : Electrical Energy Conservation & Management

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The unit of illumination is (CO1)
 a) Lux b) Luminaire
 c) Lumens d) None of the above
 Q.2 Luminous efficiency of which of the following is the highest (CO1)
 a) CFL b) HPMV
 c) HPSV d) LPSV
 Q.3 The main energy management strategies (CO2)
 a) Energy development
 b) Energy Conservation
 c) Energy scheduling
 d) Energy efficiency
 Q.4 Magnetic core losses arises due to (CO4)
 a) Eddy currents b) Hysteresis losses
 c) Magnetic saturation d) All of these

- Q.5 CFL stands for _____. (CO6)
- Compact Fluorescent lamp
 - Constantan filament lamp
 - Closed filament lamp
 - Closed fluorescent lamp
- Q.6 BIS stands for _____. (CO7)
- Board of Indian standards
 - Bureau of India standards
 - Bureau of international specifications
 - Board of international standards
- Q.7 BEE is Established on _____. (CO7)
- 1st March 2001
 - 1st March 2002
 - 1st March 2005
 - 1st March 2018
- Q.8 Which of the following is not a method of power factor corrections. (CO5)
- Auto control
 - Collective correction
 - Proper Capacitor selection
 - Manual control
- Q.9 The unit of solid angle is (CO2)
- Solid angle
 - Radian
 - Candela
 - Steradian
- Q.10 Standard efficiency motor can be replaced by energy efficient motor to reduce (CO4)
- Initial cost
 - Running cost
 - Capital cost
 - Life cycle cost

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Name any two energy efficient lamps. (CO6)
- Q.12 Lumens = Candle power X _____. (CO1)
- Q.13 Name any two losses in motor. (CO3)
- Q.14 What do you mean by lumen output? (CO1)
- Q.15 What is specific energy consumption? (CO2)
- Q.16 Energy efficient motors reduce _____. (CO4)
- Q.17 BIS stands for _____. (CO7)
- Q.18 The full form of ECBC is _____. (CO7)
- Q.19 Define lumen per watt. (CO1)
- Q.20 The BEE star rating is a measure of _____ of an appliance. (CO7)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 What is energy management? What is the need of energy conservation? (CO2)
- Q.22 What are the benefits involved for ECBC? (CO7)
- Q.23 What are the general energy saving tips in Room Air conditioners? (CO6)
- Q.24 Discuss in detail about various type of energy efficient lamps. (CO6)
- Q.25 What are the objectives of ECBC? (CO7)
- Q.26 Write importance of energy conservation. (CO2)